The Study Notes Collection

Free Exam Prep Resources for 10 Core Subjects

Empire Rewards — Free Edition

INTRODUCTION

Research from the Association for Psychological Science shows that students who use organized study notes score **15–25% higher on exams** compared to those who only re-read textbooks. The act of reviewing structured notes activates active recall — the most effective learning technique identified by cognitive science.

Most students waste hours creating notes from scratch, often missing key concepts or including irrelevant information. This collection was compiled by top-performing students and experienced educators who understand exactly what examiners look for.

This guide gives you the framework. The actual subject notes, cheat sheets, and practice questions are in the downloadable files.

But before you dive into the subject-specific material, you need to know **how** to study. That's what this guide covers: the proven techniques, the 4-week revision plan, and the exam strategies that turn notes into grades.

Let's begin.

PART 1: WHY MOST STUDYING FAILS

The average student studies 15–20 hours per week. But most of that time is spent inefficiently.

The 3 Most Common Study Mistakes

Mistake 1: Re-reading notes and textbooks.
Re-reading feels productive because you're moving your eyes across text. But research consistently shows that re-reading produces almost zero long-term retention. Your brain treats it as familiar, not learned.

Mistake 2: Highlighting.
Same problem. Highlighting makes you feel like you're engaging with the material, but you're not forcing your brain to retrieve anything. Passive review doesn't create memory.

Mistake 3: Cramming the night before.
Cramming works for a test the next morning. It fails for anything you actually need to remember. Spaced repetition — reviewing material at increasing intervals — can improve long-term retention by up to 200% compared to massed practice.

The Fix

Active recall + spaced repetition. These two techniques are the foundation of every study method that actually works.

PART 2: THE 4 STUDY TECHNIQUES THAT WORK

These are backed by cognitive science research. Use them consistently and you'll retain more information in less time.

Technique 1: Active Recall

What it is: Instead of re-reading, close your notes and try to write down everything you remember. Then check what you missed.

Why it works: The act of retrieving information from memory strengthens the neural pathways that store it. This is called the "testing effect," and it's the single most effective study technique identified by research.

How to do it:

1. Read a section of notes
2. Close the notes
3. Write down everything you remember on a blank page
4. Open the notes and check what you missed
5. Repeat the missed items

Time: 15 minutes per topic

Technique 2: Spaced Repetition

What it is: Review material at increasing intervals rather than all at once.

Why it works: Your brain consolidates memories during sleep and downtime. Reviewing material at 1 day, 3 days, 7 days, and 14 days leverages the spacing effect and improves long-term retention by up to 200%.

How to do it:

- Day 1: Learn the material
- Day 2: Review (10 min)
- Day 5: Review (5 min)
- Day 12: Review (3 min)
- Day 26: Review (2 min)

Each review takes less time than the one before because the memory is stronger.

Time: 20 minutes total across 4 weeks per topic

Technique 3: The Cornell Method

What it is: A note-taking format that divides the page into three sections: notes, cues, and summary.

Why it works: The cue column forces you to actively process what you're writing during lectures. The summary section forces you to synthesize at the end. Both activate recall.

How to do it:

- Divide your page: 2/3 for notes (right), 1/3 for cues (left), bottom for summary
- During lecture: Take notes in the right column
- After lecture: Fill in cues (questions) in the left column
- End of day: Write a 2–3 sentence summary at the bottom
- Before exam: Cover the notes, use cues to quiz yourself

Time: Same as regular note-taking, plus 5 min of processing after

Technique 4: The Feynman Technique

What it is: Explain the concept out loud as if teaching it to a 12-year-old.

Why it works: The act of explaining forces you to find gaps in your understanding. When you get stuck, you've found exactly what you don't know.

How to do it:

1. Pick a concept
2. Write the concept name at the top of a blank page
3. Explain it in plain language, as if to a beginner
4. When you get stuck, go back to notes, fill the gap, then continue

5. Simplify further if needed

   Time: 10 minutes per concept

PART 3: THE 4-WEEK REVISION PLAN

This plan assumes you have 4 weeks before exams. Adjust the timeline if you have more or less time.

Week 1: Review All Notes

Goal: Refresh everything once. Identify weak topics.

Daily time: 1.5–2 hours

Structure:

- Morning (30 min): Review yesterday's notes using active recall
- Midday (60 min): Read through new topic notes
- Evening (30 min): Write a summary of the day's topic in your own words

End of week: List the 5 weakest topics you found.

Week 2: Deep Dive on Weak Topics

Goal: Master the concepts you struggled with in Week 1.

Daily time: 2–3 hours

Structure:

- Morning (45 min): Spaced repetition on the previous day's topic
- Midday (90 min): Deep study on a weak topic — read, take notes, explain out loud
- Evening (30 min): Practice questions on the weak topic

End of week: All weak topics should feel significantly clearer.

Week 3: Practice Exams

Goal: Build exam-taking stamina and time management.

Daily time: 3–4 hours

Structure:

- Morning (60 min): Timed practice exam section
- Midday (60 min): Review mistakes — write down every mistake and why
- Afternoon (60 min): Additional practice questions on the topic
- Evening (30 min): Active recall on the day's material

End of week: You should be scoring 10–20% higher than Week 2.

Week 4: Final Review

Goal: Consolidate. Don't cram new material.

Daily time: 2–3 hours

Structure:

- Morning (60 min): Quick review of all cheat sheets
- Midday (60 min): Practice a few questions from each subject
- Afternoon (30 min): Light review of tough concepts
- Evening (30 min): Rest. No new material.

Day before exam: No studying. Eat well, sleep 8 hours, prepare your materials.

PART 4: THE CHEAT SHEET LIBRARY

Here's a preview of the formula cheat sheets included in the download.

Physics — Kinematics

Constant acceleration equations:

```
text
v = v₀ + at
x = x₀ + v₀t + ½at²
v² = v₀² + 2a(x − x₀)
x = x₀ + ½(v₀ + v)t
```

Where: v = final velocity, v_0 = initial velocity, a = acceleration, t = time, x = final position, x_0 = initial position

Mathematics — Calculus

Key derivative rules:

```
text
d/dx (xⁿ) = nxⁿ⁻¹
d/dx (sin x) = cos x
d/dx (cos x) = −sin x
d/dx (eˣ) = eˣ
d/dx (ln x) = 1/x
Product rule: (uv)' = u'v + uv'
Quotient rule: (u/v)' = (u'v − uv')/v²
Chain rule: d/dx f(g(x)) = f'(g(x)) · g'(x)
```

Chemistry — Stoichiometry

Key formulas:

```
text
Moles = mass / molar mass
Moles = concentration × volume (in liters)
Ideal gas law: PV = nRT
% yield = (actual yield / theoretical yield) × 100
Molarity (M) = moles of solute / liters of solution
```

Biology — Cellular Respiration

The three stages:

```
text
1. Glycolysis (cytoplasm): Glucose → 2 Pyruvate + 2 ATP + 2 NADH
2. Krebs Cycle (mitochondrial matrix): 2 Acetyl-CoA → 6 NADH + 2 FADH₂ + 2 ATP + 4 CO₂
3. Electron Transport Chain (inner mitochondrial membrane): ~32 ATP
```

Total ATP yield: ~36–38 ATP per glucose molecule

Economics — Key Formulas

```
text
GDP = C + I + G + (X − M)
Inflation rate = ((CPI₂ − CPI₁) / CPI₁) × 100
Unemployment rate = (Unemployed / Labor Force) × 100
Price elasticity of demand = %Δ Quantity / %Δ Price
```

Psychology — Key Theories

```
text
Classical Conditioning (Pavlov): US → UR; US + CS → UR; CS → CR
Operant Conditioning (Skinner): Reinforcement increases behavior; punishment decreases behavior
Maslow's Hierarchy: Physiological → Safety → Love/Belonging → Esteem → Self-Actualization
Piaget's Stages: Sensorimotor (0–2) → Preoperational (2–7) → Concrete Operational (7–11) → Formal Operational (11+)
```

The full cheat sheets in the download include hundreds more formulas across all 10 subjects, each organized by topic with units and common values.

PART 5: SAMPLE PRACTICE QUESTIONS

Here are samples from each subject's practice question set. The full pack includes 200+ questions with detailed answers and explanations.

Mathematics Sample

Question: Find the derivative of $f(x) = 3x^2 + 2x - 5$.

Answer: $f'(x) = 6x + 2$

Explanation: Apply the power rule to each term. $d/dx(3x^2) = 6x$. $d/dx(2x) = 2$. $d/dx(-5) = 0$.

Biology Sample

Question: During which phase of mitosis do chromosomes align at the cell's equator?

Answer: Metaphase

Explanation: In metaphase, spindle fibers attach to centromeres and pull chromosomes to the metaphase plate (the equator of the cell). This is followed by anaphase, when sister chromatids separate.

Chemistry Sample

Question: How many moles are in 44 grams of CO_2?

Answer: 1 mole

Explanation: Molar mass of CO_2 = 12 (C) + 32 (O_2) = 44 g/mol. Moles = mass / molar mass = 44 / 44 = 1 mole.

Physics Sample

Question: A ball is dropped from 20 meters. How long does it take to hit the ground? (Assume g = 10 m/s^2.)

Answer: 2 seconds

Explanation: Use $x = x_0 + v_0t + \frac{1}{2}at^2$. Set x = 0, x_0 = 20, v_0 = 0, a = −10. Solve $0 = 20 - 5t^2$. $t^2 = 4$. t = 2 seconds.

Economics Sample

Question: If CPI rises from 120 to 126, what is the inflation rate?

Answer: 5%

Explanation: Inflation rate = ((126 − 120) / 120) × 100 = 5%.

Psychology Sample

Question: A child learns to associate the sound of the ice cream truck with getting ice cream. What type of conditioning is this?

Answer: Classical conditioning

Explanation: The ice cream truck sound (initially neutral) becomes a conditioned stimulus through repeated pairing with the unconditioned stimulus (ice cream).

PART 6: THE 4-WEEK REVISION TIMETABLE PLANNER

The downloadable planner is a spreadsheet that lets you customize the entire 4-week revision schedule. Here's the structure.

The Planner Includes:

- **Daily time blocks:** Morning, midday, evening slots for each subject
- **Subject rotation:** Ensures you don't neglect any subject for more than 2 days

- **Progress tracking:** Checkboxes for each topic completed
- **Weak topic log:** A section for identifying and revisiting weak areas
- **Pomodoro timers:** Suggested 25-minute blocks with 5-minute breaks
- **Rest day scheduling:** Built-in days off to prevent burnout

How to Use It

1. List every subject and every topic within each subject
2. Assign topics to weeks based on priority (weakest first)
3. Fill in the daily time blocks with subject and topic
4. Check off each topic as you complete it
5. At the end of each week, review progress and adjust

Sample Week

Day	Morning	Midday	Evening
Mon	Math: Calculus review	Biology: Genetics	Math: Practice problems
Tue	Biology: Cell biology	Chemistry: Stoichiometry	Bio: Practice questions
Wed	Chemistry: Organic chem	Physics: Kinematics	Chem: Practice problems
Thu	Physics: Mechanics	Math: Statistics	Physics: Practice questions
Fri	Math: Linear algebra	Economics: Micro	Econ: Practice problems
Sat	Review weak topics	Practice exam section	Rest
Sun	Rest	Light review of the week	Plan next week

PART 7: EXAM DAY STRATEGIES

Knowing the material is only half the battle. Here's how to perform on exam day.

The Night Before

- **No new studying.** Cramming the night before increases anxiety and reduces sleep quality.
- **Prepare your materials.** ID, pens, calculator (if allowed), water bottle.
- **Sleep 7–8 hours.** Sleep consolidates memory. Cutting sleep to study more is self-defeating.

Exam Morning

- **Eat a real breakfast.** Protein and complex carbs, not just coffee.
- **Arrive early.** Rushing increases anxiety.
- **Skip the last-minute cram.** Other students' panic is contagious.

During the Exam

First 2 minutes: Skim the entire exam. Identify the easiest questions and the hardest ones.

Strategy: Answer the easiest questions first. This banks points and builds confidence.

Time management: Divide the exam time by the number of questions. If you have 60 minutes and 30 questions, you have 2 minutes per question. Stick to the pace.

When stuck: Skip and come back. Don't lose 10 minutes on one hard question.

Multiple-choice strategy:

- Read the question twice
- Eliminate obviously wrong answers
- If two answers look correct, the more specific one is usually right
- Trust your first instinct — don't change answers unless you're certain

Essay strategy:

- Read the prompt carefully
- Outline before writing (3–5 minutes)
- Start with a clear thesis statement
- One paragraph per argument
- Conclude with a summary, not new information

Math/science strategy:

- Write down what you know (given values)
- Write down what you need to find
- Identify the relevant formula
- Show your work — partial credit adds up
- Check units at the end

After the Exam

- **Don't discuss answers with classmates.** It only increases anxiety for the next exam.
- **Move on.** What's done is done. Focus on the next exam.

PART 8: THE SUBJECT-BY-SUBJECT STUDY GUIDE

Here's what to focus on for each of the 10 subjects in this collection.

Mathematics

Focus on: understanding concepts, not memorizing steps. Practice problems are more valuable than re-reading formulas. Identify the 5–10 problem types most likely to appear and master those.

Biology

Focus on: processes and relationships, not isolated facts. Use diagrams to understand how systems connect. Know the key experiments and what they demonstrated.

Chemistry

Focus on: reaction mechanisms and stoichiometry. Practice balancing equations until it's automatic. Know the periodic table trends.

Physics

Focus on: setting up problems correctly. Identify given values, unknowns, and relevant formula. Units are clues — if the answer requires m/s, you need velocity.

Economics

Focus on: graph interpretation and real-world applications. Know the key models (supply and demand, AD-AS, Phillips curve) and when to use them.

Psychology

Focus on: key theories and supporting research. For each theory, know the theorist, the core idea, and one supporting study.

Computer Science

Focus on: algorithms and data structures. Understand time complexity. Practice writing pseudocode for common operations.

Business Studies

Focus on: case study analysis. Know the frameworks (SWOT, PESTLE, Porter's Five Forces) and when to apply them.

English Literature

Focus on: themes, characters, and literary devices. Know the key quotes for each text. Practice writing analytical paragraphs.

History

Focus on: cause and effect. Know the timeline of major events. Practice evaluating primary sources.

FREQUENTLY ASKED QUESTIONS

What subjects are covered?

Mathematics, Biology, Chemistry, Physics, Economics, Psychology, Computer Science, Business Studies, English Literature, and History.

How many practice questions are included?

Over 200 questions with detailed answers and explanations, organized by subject and topic.

Are the study notes suitable for college students?

Yes — they're designed for college and university-level courses. High school students taking AP or IB courses will also find them useful.

What's in the revision timetable planner?

A customizable spreadsheet with daily planner, weekly overview, topic tracker, progress checklist, and Pomodoro timer integration.

Can I print the cheat sheets?

Yes. Every cheat sheet fits on a single page and is designed for printing.

How long before my exam should I start using these notes?

2–3 weeks before for best results. Even if your exam is tomorrow, the cheat sheets and practice questions can help you focus.

Do I need special software?

No. The download is a ZIP file with PDFs and spreadsheets. Any PDF reader and any spreadsheet app will work.

CONCLUSION

You now have:

- The 4 study techniques that actually work
- A complete 4-week revision plan
- Cheat sheets across all 10 subjects
- 200+ practice questions with answers
- A customizable revision timetable planner
- Exam day strategies
- Subject-specific study guidance

The only thing left is to start. Pick your weakest subject. Open the notes. Do the first practice question. Then another. Then another.

Consistency over 4 weeks beats intensity over 4 days. Every time.

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